❸ Home Tracker: Construction Manager (CM) and Debris and Demolition Manager (DDM)

Instructions:

- Put a "B" when it is built, "S" when in stock, and "D" when it is demolished. Write the *B year* and *D year* and calculate the *Lifespan = D year – B year*.
- At the bottom, put the count of SF and MF homes Built (B), Demolished (D) that year.
- For Stock (S) *calc*: Calculate the $Stock_t = Stock_{t-1} + \sum FlowIn_t - \sum FlowOut_t$. Note that $Stock_{1999} = 0$.
- For Stock (S) *count:* put the count of SF and MF homes that are new or already existing Stock (S) that year. Compare your count to the calculation.

Address	Type	2000	2010	2020	2030	2040	2050	2060	2070	2080	B year	D year	Lifespan
1 Sweet Street	SF	B	S	S	S	S	S	D			2000	2060	60
2 Sweet Street	SF		B	S	S	D					2010	2040	30
3 Sweet Street	SF			B	S	S	S	D			2020	2060	40
4 Sweet Street	SF			B	S	D					2020	2040	20
5 Sweet Street	SF				B	S	S	D			2030	2060	30
6 Sweet Street	SF				B	S	S	D			2030	2060	30
1-3 Sugar Ave	MF				B	S	S	S	S	D	2030	2080	50
5-7 Sugar Ave	MF				B	S	S	S	S	D	2030	2080	50
2-4 Sugar Ave	MF					B	S	S	S	D	2040	2080	40
6-8 Sugar Ave	MF					B	S	S	S	D	2040	2080	40

		2000	2010	2020	2030	2040	2050	2060	2070	2080
Built (B)	SF	1	1	2	2	0	0	0	0	0
Demolish (D)	SF	0	0	0	0	2	0	4	0	0
Stock (S) *calc.*	SF	1	2	4	6	4	4	0	0	0
Stock (S) *count*	SF	1	2	4	6	4	4	0	0	0
Built (B)	MF	0	0	0	2	2	0	0	0	0
Demolish (D)	MF	0	0	0	0	0	0	0	0	4
Stock (S) *calc.*	MF	0	0	0	2	4	4	4	4	0
Stock (S) *count*	MF	0	0	0	2	4	4	4	4	0
Stock (S) *sum*	Total	1	2	4	8	8	8	4	4	0

❶ Flow Tracker: Materials Warehouse Manager (MWM)

Instructions:

- Flow In_{2000} = initial new materials
- Flow $In_{>2000}$ = purchase of recycled materials from WRM
- Flow Out = sales of materials to CRM
- $Stock_t = Stock_{t-1} + \sum FlowIn_t - \sum FlowOut_t$
  - rearrange the above formula to calculate stock change based on Flow In and Flow Out
- Stock Δ = stock change = $Stock_t - Stock_{t-1}$
- $Stock_{1999} = 0$

	Brick				Wood				Metal				Glass			
Year	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock
2000	84	3	81	81	24	3	21	21	24	1	23	23	24	1	23	23
2010	0	3	-3	78	0	3	-3	18	0	1	-1	22	0	1	-1	22
2020	0	6	-6	72	0	6	-6	12	0	2	-2	20	0	2	-2	20
2030	0	34	-34	38	0	6	-6	6	0	10	-10	10	0	6	-6	14
2040	2	28	-26	12	4	0	4	10	2	8	-6	4	5	4	1	15
2050	0	0	0	12	0	0	0	10	0	0	0	4	0	0	0	15
2060	4	0	4	16	8	0	8	18	4	0	4	8	10	0	10	25
2070	0	0	0	16	0	0	0	18	0	0	0	8	0	0	0	25
2080	24	0	24	40	0	0	0	18	8	0	8	16	40	0	40	65
Total	114	74	40	N/A	36	18	18	N/A	38	22	16	N/A	79	14	65	N/A

❻ Flow Tracker: Waste & Recycling Manager (WRM)

Instructions:

- Flow In = scrap and debris from DDM
- Flow Out = sales of reusable or recycled materials to MWM
- $Stock_t = Stock_{t-1} + \sum FlowIn_t - \sum FlowOut_t$
    - rearrange the above formula to calculate stock change based on Flow In and Flow Out
- Stock Δ = stock change = $Stock_t - Stock_{t-1}$
- $Stock_{1999} = 0$

	Brick for Reuse				Wood for Reuse				Metal for Recycling				Combined Scrap & Debris for Landfill			
Year	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock
2000	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2010	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2020	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2030	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2040	2	2	0	0	4	4	0	0	2	2	0	0	5	5	0	0
2050	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2060	4	4	0	0	8	8	0	0	4	4	0	0	10	10	0	0
2070	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2080	24	24	0	0	0	0	0	0	8	8	0	0	40	40	0	0
Total	30	30	0	N/A	12	12	0	N/A	14	14	0	N/A	55	55	0	N/A

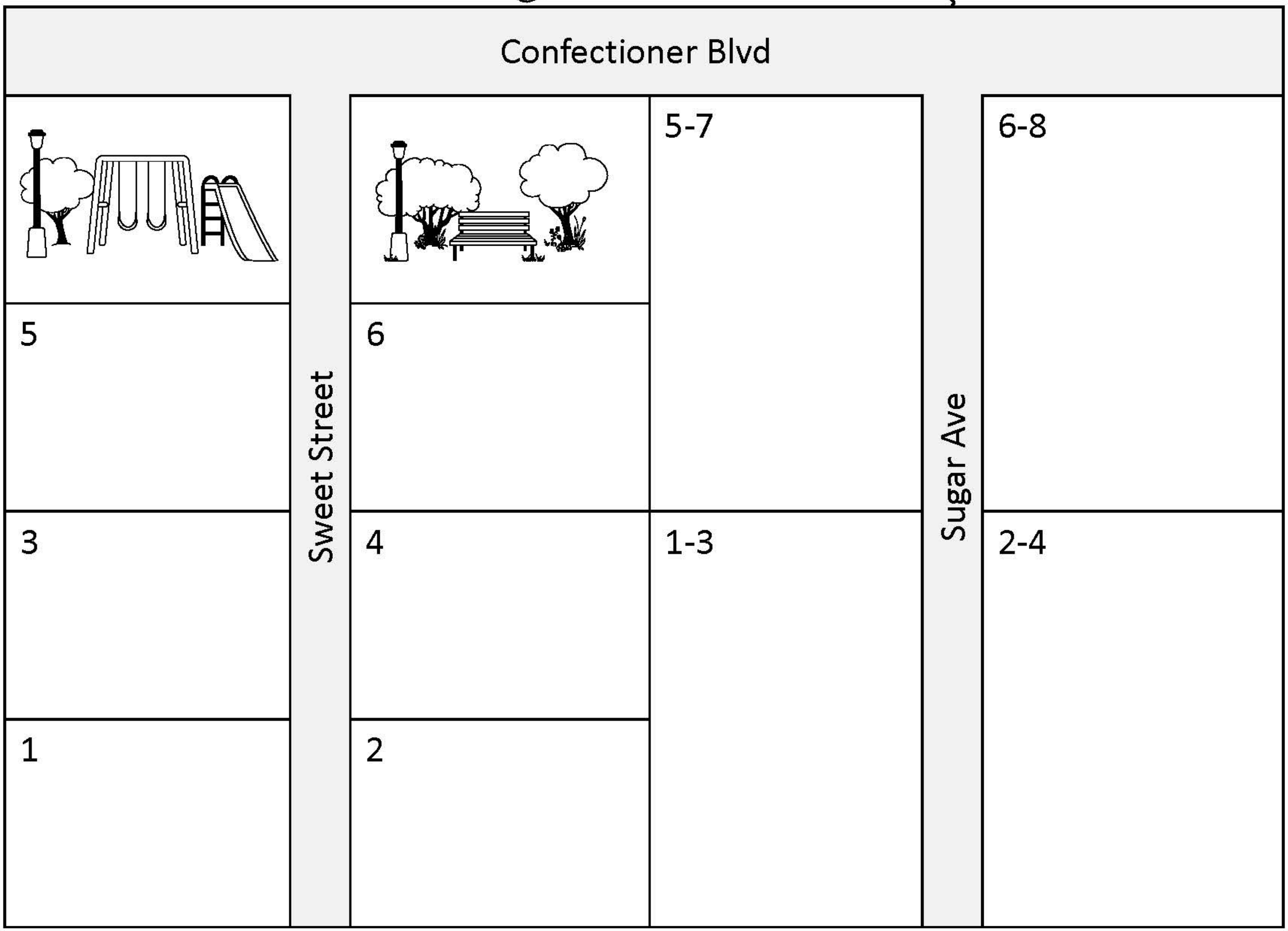
❹ Flow Tracker: Construction Manager (CM) and Debris and Demolition Manager (DDM)

Instructions:

- Flow In = construction materials from MWM
- Flow Out = scrap and debris to WRM
- $Stock_t = Stock_{t-1} + \sum FlowIn_t - \sum FlowOut_t$
  - rearrange the above formula to calculate stock change based on Flow In and Flow Out
- Stock Δ = stock change = $Stock_t - Stock_{t-1}$
- $Stock_{1999} = 0$
- For the 50th Anniversary Golden Jubilee, put the count in the specified row for comparison.

	Brick				Wood				Metal				Glass			
Year	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock
2000	3	1	2	2	3	1	2.5	2.5	1	0	1	1	1	0	1	1
2010	3	1	2	4	3	1	2.5	5	1	0	1	2	1	0	1	2
2020	6	2	4	8	6	1	5	10	2	0	2	4	2	0	2	4
2030	34	6	28	36	6	1	5	15	10	0	10	14	6	0	6	10
2040	28	8	20	56	0	5	-5	10	8	2	6	20	4	2	2	12
2050	0	0	0	56	0	0	0	10	0	0	0	20	0	0	0	12
2050	Golden Jubilee Count:			56	Golden Jubilee Count:			10	Golden Jubilee Count:			20	Golden Jubilee Count:			12
2060	0	8	-8	48	0	10	-10	0	0	4	-4	16	0	4	-4	8
2070	0	0	0	48	0	0	0	0	0	0	0	16	0	0	0	8
2080	0	48	-48	0	0	0	0	0	0	16	-16	0	0	8	-8	0
Total	74	74	0	N/A	18	18	0	N/A	22	22	0	N/A	14	14	0	N/A

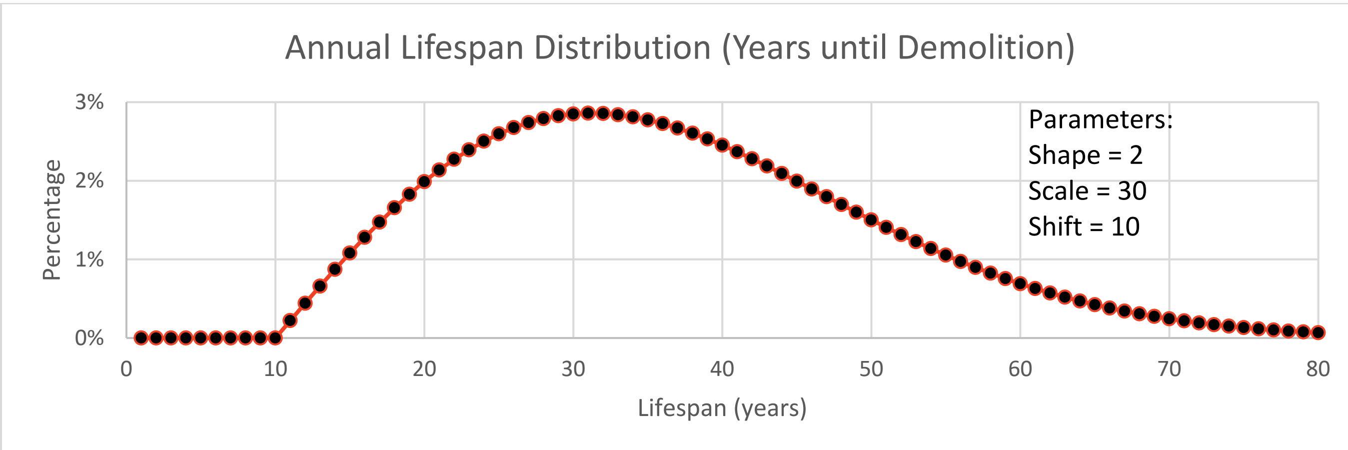
Candy Town Map



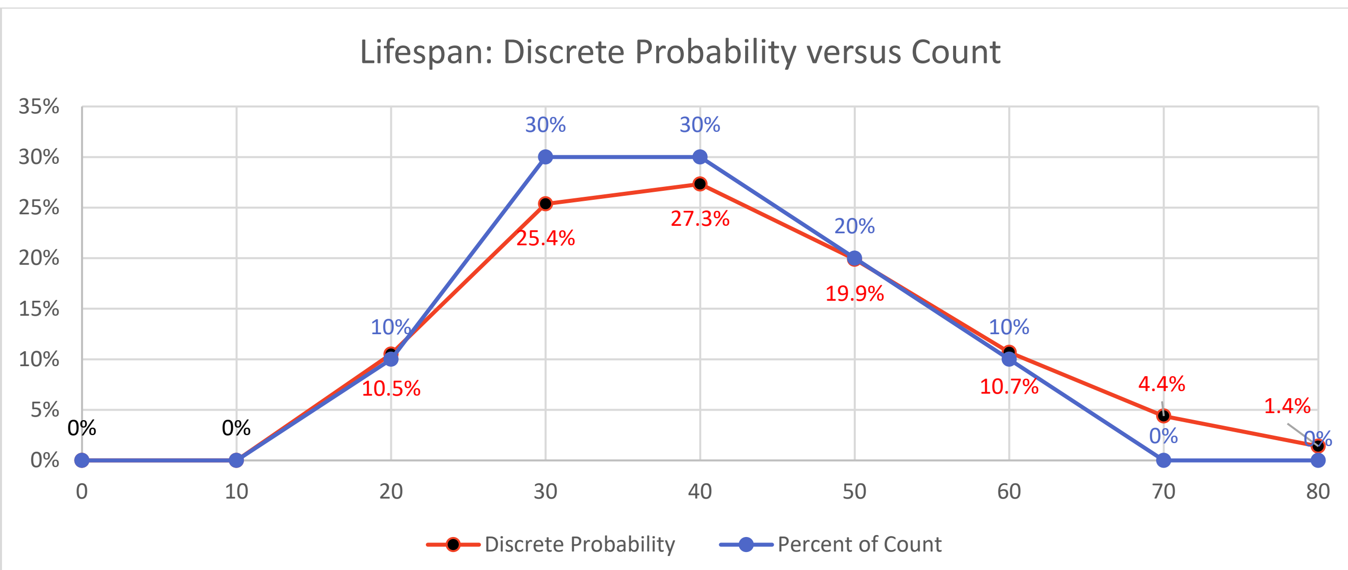
❺ Lifespan Distribution

Industrial Ecology researchers who recently settled in Hummus Town realized that the buildings in Candy Town could make an excellent case study. They modeled the lifespan distribution of the buildings using a Weibull Distribution, which is a commonly used approach for lifespans, since it was developed with that sort of purpose in mind. Instead of mean and standard deviation, the parameters for a Weibull Distribution are shape, scale, (and optionally shift).

Each point in the chart below represents the likelihood a house would be demolished after that many years.



Let's compare this to the data you collected. Since data was only available in Candy Town every decade, the researchers took the difference between *cumulative* Weibull distribution above over the preceding range. For instance, the 10.5% is the difference between the cumulative distribution at year 20 and year 10; 25.4% is the value at year 30 minus year 20.



Based on your results in the Home Tracker, fill out the table below, and plot your percentages above!

- Count how many buildings have each lifespan (years).
- Find the percentage of buildings with that lifespan, noting there are 10 buildings total.
- Compare your results to the model estimates, plotted above, and in the bottom row of the table.

Lifespan	**0**	**10**	**20**	**30**	**40**	**50**	**60**	**70**	**80**
Count	*0*	*0*	*1*	*3*	*3*	*2*	*1*	*0*	*0*
Percent	*0%*	*0%*	*10%*	*30%*	*30%*	*20%*	*10%*	*0%*	*0%*
Model	0.0%	0.0%	10.5%	25.4%	27.3%	19.9%	10.7%	4.4%	1.4%

2 Materials Warehouse Manager (MWM): Keep your materials here

Brick	Wood
Metal	**Glass**

❼ Waste & Recycling Manager (WRM): Keep your materials here

Brick for Reuse	Wood for Reuse
Metal for Recycling	**Combined Scrap & Debris for Landfill**